

HW 6: Relationship between Confidence Intervals & Significance Tests

1) Complete the following table, so the information agrees across the row.

<i>Confidence Interval</i>	<i>Hypotheses and P-Value</i>	<i>Conclusion</i>
$x = 15$ $n = 50$ C-Level: 0.95 _____	$H_0: p = 0.39$ $H_a: p \neq 0.39$ p-value = _____	
$x = \underline{\hspace{2cm}}$ $n = 150$ C-Level: 0.90 (0.5342, 0.6658)	$H_0: p = 0.68$ $H_a: \underline{\hspace{2cm}}$ p-value = _____	
$x = 85$ $n = 105$ C-Level: _____ _____	$H_0: \underline{\hspace{2cm}}$ $H_a: p \neq 0.73$ p-value = _____	We reject H_0 at $\alpha = 0.08$

2) Is the ratio of males and females that attend a large university even? An SRS of 254 students that attend the college is taken and 145 of them were female. A 99% confidence interval for p = the proportion of females that attend the university is given by (0.491, 0.651)

a) What would be the appropriate hypotheses for this problem?

b) If you wanted to use this confidence interval to draw a conclusion about a two-sided significance test, what significance level would you need to use?

c) Use the confidence interval to determine whether this test would reject or fail to reject the null hypothesis. Explain your reasoning.

d) What information is provided by the confidence interval that would not be provided by a test of significance alone?

3) In political campaigns, an attack ad (“mudslinging”) is an advertisement whose message is designed to personally attack an opposing candidate in order to gain support for the candidate waging the attack. A recent candidate running for governor noticed his support in the polls had slipped, and only 45% of voters now supported him. His campaign ran the attack ad, hoping to boost his voter approval. Two weeks after the ad ran, the poll was run again and it found that out of the SRS of 450 voters, 233 supported the candidate. The 95% confidence interval for the proportion of voters who support him is (0.472, 0.564).

a) Why is a two-sided hypothesis test appropriate to run in this situation?

b) Suppose his pollsters conducted a test of $H_0: p = 0.45$ vs the alternative $H_a: p \neq 0.45$, using $\alpha = 0.05$. Use the confidence interval to determine whether this test would reject or fail to reject the null hypothesis. Explain your reasoning.

c) Find the P-value for the test described in part (a), and explain what it measures in the context of the problem.